

Formulas:

Gaussian factors:

Canonical form:

$$\mathcal{C}(\mathbf{x}; \mathbf{K}, \mathbf{h}, g) = \exp \left\{ -\frac{1}{2} \mathbf{x}^T \mathbf{K} \mathbf{x} + \mathbf{h}^T \mathbf{x} + g \right\}$$

Mean-covariance form:

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Conversions between forms:

$$\begin{aligned} \mathbf{K} &= \boldsymbol{\Sigma}^{-1}; & \mathbf{h} &= \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}; & g &= -\frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} - \log((2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}) \\ \boldsymbol{\Sigma} &= \mathbf{K}^{-1}; & \boldsymbol{\mu} &= \mathbf{K}^{-1} \mathbf{h} \end{aligned}$$

Multiplication:

$$\mathcal{C}(\mathbf{x}; \mathbf{K}_1, \mathbf{h}_1, g_1) \cdot \mathcal{C}(\mathbf{x}; \mathbf{K}_2, \mathbf{h}_2, g_2) = \mathcal{C}(\mathbf{x}; \mathbf{K}_1 + \mathbf{K}_2, \mathbf{h}_1 + \mathbf{h}_2, g_1 + g_2)$$

Division:

$$\frac{\mathcal{C}(\mathbf{x}; \mathbf{K}_1, \mathbf{h}_1, g_1)}{\mathcal{C}(\mathbf{x}; \mathbf{K}_2, \mathbf{h}_2, g_2)} = \mathcal{C}(\mathbf{x}; \mathbf{K}_1 - \mathbf{K}_2, \mathbf{h}_1 - \mathbf{h}_2, g_1 - g_2)$$

Marginalisation:

$$\begin{aligned} \int \mathcal{C} \left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}; \begin{bmatrix} \mathbf{K}_{xx} & \mathbf{K}_{xy} \\ \mathbf{K}_{yx} & \mathbf{K}_{yy} \end{bmatrix}, \begin{bmatrix} \mathbf{h}_x \\ \mathbf{h}_y \end{bmatrix}, g \right) d\mathbf{y} &= \mathcal{C}(\mathbf{x}; \mathbf{K}', \mathbf{h}', g'), \text{ where} \\ \mathbf{K}' &= \mathbf{K}_{xx} - \mathbf{K}_{xy} \mathbf{K}_{yy}^{-1} \mathbf{K}_{yx} \\ \mathbf{h}' &= \mathbf{h}_x - \mathbf{K}_{xy} \mathbf{K}_{yy}^{-1} \mathbf{h}_y \\ g' &= g + \frac{1}{2} \left(\log |2\pi \mathbf{K}_{yy}^{-1}| + \mathbf{h}_y^T \mathbf{K}_{yy}^{-1} \mathbf{h}_y \right) \end{aligned}$$

Observe and reduce:

$$\begin{aligned} \mathcal{C} \left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} = \hat{\mathbf{y}} \end{bmatrix}; \begin{bmatrix} \mathbf{K}_{xx} & \mathbf{K}_{xy} \\ \mathbf{K}_{yx} & \mathbf{K}_{yy} \end{bmatrix}, \begin{bmatrix} \mathbf{h}_x \\ \mathbf{h}_y \end{bmatrix}, g \right) &= \mathcal{C}(\mathbf{x}; \mathbf{K}', \mathbf{h}', g'), \text{ where} \\ \mathbf{K}' &= \mathbf{K}_{xx} \\ \mathbf{h}' &= \mathbf{h}_x - \mathbf{K}_{xy} \hat{\mathbf{y}} \\ g' &= g + \mathbf{h}_y^T \hat{\mathbf{y}} - \frac{1}{2} \hat{\mathbf{y}}^T \mathbf{K}_{yy} \hat{\mathbf{y}} \end{aligned}$$

Linear Gaussian:

$$\begin{aligned} \mathbf{y} &= \mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n}, & \mathbf{n} &\sim \mathcal{N}(\mathbf{n}; \mathbf{0}, \boldsymbol{\Sigma}_{\mathbf{n}}) \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{C} \left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}; \mathbf{K}_{\mathbf{y}|\mathbf{x}}, \mathbf{h}_{\mathbf{y}|\mathbf{x}}, g_{\mathbf{y}|\mathbf{x}} \right), \text{ where} \\ \mathbf{K}_{\mathbf{y}|\mathbf{x}} &= \begin{bmatrix} \mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ -\boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \\ \mathbf{h}_{\mathbf{y}|\mathbf{x}} &= \begin{bmatrix} -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \mathbf{b} \\ g_{\mathbf{y}|\mathbf{x}} &= -\frac{1}{2} \mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{b} - \log \left((2\pi)^{\frac{k}{2}} |\boldsymbol{\Sigma}_{\mathbf{n}}|^{\frac{1}{2}} \right) \end{aligned}$$

Formulas continue on the next page

Parameter estimation:

ML estimation:

$$\boldsymbol{\Theta}^{\text{ML}} = \arg \max_{\boldsymbol{\Theta}} \log p(\mathcal{D} | \boldsymbol{\Theta})$$

MAP estimation:

$$\boldsymbol{\Theta}^{\text{MAP}} = \arg \max_{\boldsymbol{\Theta}} \log p(\boldsymbol{\Theta} | \mathcal{D})$$

The EM algorithm:

Algorithm 1 The expectation-maximisation (EM) algorithm

Initialise $\boldsymbol{\Theta}^{\text{old}}$

while not converged **do**

$$q(\mathcal{H}) \leftarrow p(\mathcal{H} | \mathcal{D}, \boldsymbol{\Theta}^{\text{old}})$$

▷ E-step

$$\boldsymbol{\Theta}^{\text{new}} \leftarrow \arg \max_{\boldsymbol{\Theta}} Q(\boldsymbol{\Theta}), \quad \text{where } Q(\boldsymbol{\Theta}) = \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H} | \boldsymbol{\Theta})$$

▷ M-step

$$\boldsymbol{\Theta}^{\text{old}} \leftarrow \boldsymbol{\Theta}^{\text{new}}$$

end while
